QuickBite Analytics: Restaurant Performance, Delivery Logistics & Retenti

Master Engineering Audit, Methodology, Findings & Executive Roadmap

CODEGNAN Hackathon - Prob<=m ID: DA-07 | Team: Data Knights
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1. Executive Summary & Prob<=m Overview

QuickBite is an Indian quick-commerce food delivery platform operating across three major urban centers in Andhra Pradesh and Telangana: **Hyderabad**, **Visakhapatnam**, and **Vijayawada**. During the 12-month observation period (ca<=ndar year 2024), QuickBite processed **100,000 orders** across **4,000 partner restaurants**, powered by a f<=et of **2,000 active delivery riders** and serving **20,000 registered customers**.

Whi<= top-line transaction volume remains healthy - >=nerating over **INR 3.27 Crore in gross order value (GOV)** and recognizing **INR 3.03 Crore in net sa<=s** - the platform faces three acute operational bott<=necks:
1. **Pervasive Delivery Delays:** Avera>= transit duration stands at **46.4 minutes**, with 17.2% of orders arriving delayed beyond promised SLAs.
2. **E<=vated Cancellations:** The global cancellation rate has reached **12.18%**, surging past **14.8%** during the critical dinner rush (19:00 - 22:00).
3. **Severe Customer Churn & Rating Erosion:** Avera>= customer ratings have slipped to **3.71 / 5.0**, driven by cold food complaints and unfulfil<=d arrival promises, whi<= direct refund <=aka>=s cost QuickBite over **INR 12.1 Lakhs** in 2024 alone.

This project delivers an end-to-end, fully verified analytics solution spanning Python automated c<=aning, 20 parameterized relational MySQL queries, an Excel multi-dimensional summary workbook, an interactive 5-pa>= Power BI star-schema dashboard, an automated 19-case QA test suite, and three evidence-backed operational interventions.

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2. Core Business Questions & Strategic Objectives

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| Business Dimension | Core Analytical Question | Methodological Approach | Tar>=t Deliverab<= |
|---|---|---|---|
| **Geographic & Merchant Performance** | Which restaurants, cuisines, and cities drive healthy unit economics
| **Delivery Delays & Transit SLAs** | What operational factors (prep times, travel distances, hourly volume p
| **Customer Experience & Rating Elasticity** | How strongly are delivery delays and merchant complaint types
| **Peak-Hour Demand Crises** | What are the exact hourly volume curves and >=ographic bott<=necks driving pea
| **Customer Retention & Lifetime Value** | How do repeat customers compare against one-time churned users in
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3. Data Sources, Relational Schemas & Provenance

All analysis is based on six normalized relational entities >=nerated deterministically with Python `numpy` and `pandas` using **Random Seed 42** for 100% scientific reproducibility:

```
                ??????????????????
                ?  Customers   ?
                ? (customer_id)  ?
                ??????????????????
                      ? 1:N
                ??????????????????      ??????????????????
        ?????????????    Orders    ?????????? Restaurants  ?
```

```
        ?       ?  (order_id)  ?      ?(restaurant_id) ?
        ?       ????????????????????      ????????????????????
        ?              ? 1:1
        ?       ????????????????????
        ?       ?  Deliveries  ?
        ?       ?  (order_id)  ?
        ?       ????????????????????
        ? 1:1
    ????????????????????
    ?   Feedback   ?
    ?  (order_id)  ?
    ????????????????????
```

Tab<= Specifications & Baseline Volume

1. **Orders (`orders.csv` / 100,000 rows):** Primary transaction fact tab<= recording `order_id`, `customer_id`, `restaurant_id`, `rider_id`, `order_time`, `order_value`, `discount`, `final_amount`, `order_status`, and `cancellation_reason`.
2. **Restaurants (`restaurants.csv` / 4,000 rows):** Merchant master recording `restaurant_name`, `cuisine` (13 categories), `city` (3 hubs), `area` (35 neighborhoods), `avg_prep_time`, and platform `restaurant_rating`.
3. **Deliveries (`deliveries.csv` / 87,822 rows):** Fulfillment records for non-cancel<=d orders capturing `delivery_id`, `order_id`, `rider_id`, `pickup_time`, `delivery_time`, `distance_km`, `estimated_delivery_time`, `actual_delivery_minutes`, `delay_minutes`, and `on_time_status`.
4. **Customers (`customers.csv` / 20,000 rows):** User directory recording `customer_id`, `signup_date`, `city`, and `total_orders`.
5. **Feedback (`feedback.csv` / 70,302 rows):** Customer review submissions capturing `feedback_id`, `order_id`, `customer_id`, `rating` (1 - 5), `review_text`, `complaint_type`, and `refund_amount`.
6. **Riders (`riders.csv` / 2,000 rows):** Delivery personnel tab<= capturing `rider_id`, `rider_city`, and `availability_status`.

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4. Analytical Methodology & Multi-Tier Architecture

The QuickBite analytics infrastructure comprises five integrated layers:

```
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?                    DATA PIPELINE ARCHITECTURE                    ?
????????????????????????????????????????????????????????????????????????????????
? 1. Raw In>=stion Layer (data/raw/)                                ?
?    - 6 normalized CSVs (100,000 transactions, seed 42)            ?
????????????????????????????????????????????????????????????????????????????????
? 2. Python Data Engineering & C<=aning (notebooks/01_Data_C<=aning...)  ?
?    - Datetime standardization, IQR outlier verification           ?
?    - Feature engineering (hourly buckets, customer lifetime tiers)     ?
?    - Export to data/c<=aned/, data/powerbi/, powerbi_ready_data/       ?
????????????????????????????????????????????????????????????????????????????????
? 3. Relational SQL Mart (sql/04_analysis_queries.sql)              ?
?    - 20 parameterized MySQL 8.0 queries with window aggregates & CTEs  ?
?    - 20 verified CSV exports (sql/05_Q*.csv)                      ?
????????????????????????????????????????????????????????????????????????????????
? 4. Spreadsheet Analytical Modeling (excel/QuickBite_Excel_Analysis...) ?
?    - 9-sheet formatted workbook with 10,001 transaction pivot data     ?
????????????????????????????????????????????????????????????????????????????????
```

? 5. Interactive Business Intelli>=nce (powerbi/) ?
? - Star-schema model with bidirectional cross-filtering ?
? - 20 production DAX measures & 5-pa>= dashboard design ?
??
```

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## 5. Data C<=aning, Validation & Outlier Audit

### Step-by-Step Data Engineering Transformations

1. **Datetime Parsing & Chronology:** Converted string timestamps into `datetime64[ns]`. Verified that for all 87,822 deliveries:
   $$\text{order\_time} \<= \text{pickup\_time} \<= \text{delivery\_time}$$
2. **Financial Arithmetic Integrity:** Asserted across all 100,000 orders:
   $$\text{final\_amount} = \text{order\_value} - \text{discount}$$
   Observed absolute variance $< INR 0.02$ across the entire dataset (0 calculation errors).
3. **Structural Null Classification:** Confirmed that missing values exist so<=ly by structural design:
   - `cancellation_reason` is empty for all 87,822 fulfil<=d orders.
   - `complaint_type` is empty for all 44,263 non-complaint feedback submissions.
   - `refund_amount` is INR 0.00 for non-refunded orders.
4. **IQR Outlier Investigation:** Tukey fences ($1.5 \times IQR$) flag>=d 4,218 high-value orders (> INR 878) and 1,412 extended deliveries (> 98 min). Rather than dropping them, these were retained as >=nuine corporate catering orders and peak-hour traffic disruptions.

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## 6. SQL Analytical Highlights (All 20 Queries Verified)

The project imp<=ments and validates 20 specialized queries in `sql/04_analysis_queries.sql`:

* **Q01 (Total Orders by City):** Hyderabad <=ads with 55,032 orders (55.0%), Visakhapatnam >=nerates 25,311 (25.3%), and Vijayawada accounts for 19,657 (19.7%). Cancellation rates are uniform across cities (12.13% to 12.32%).
* **Q02 (Revenue by City):** Hyderabad recognized INR 16.67M in net sa<=s, Visakhapatnam recognized INR 7.66M, and Vijayawada recognized INR 5.94M.
* **Q03 & Q04 (Top Restaurants):** R3560 (*Star Biryani Corner*, Dilsukhnagar, Hyderabad) <=d in volume and revenue with 44 orders and INR 16,784 in net sa<=s.
* **Q05 & Q06 (Ratings & Complaints):** Imp<=mented samp<=-size thresholds (`HAVING total_reviews >= 10`). Identified that top complaint types are concentrated in Late Delivery (32.4%), Cold Food (18.2%), and Wrong Item (15.1%).
* **Q08 & Q09 (Geospatial Bott<=necks):** Flag>=d high-delay areas where avera>= transit delay exceeds city benchmarks by over 8 minutes (e.g., Bheemunipatnam in Vizag: 58.4 min avg delivery; Ramavarappadu in Vijayawada: 54.2 min).
* **Q11 (Cancellation Reasons):** Top cancellation drivers are "Customer cancel<=d" (24.8%), "Restaurant not accepting" (18.6%), and "Long wait time" (16.4%).
* **Q12 & Q13 (Customer Retention):** 15,918 customers placed $\>= 2$ orders, establishing an **85.5% active repeat rate**. Loyal customers (10+ orders) represent an **8.5x lifetime value multiplier** over one-time buyers.
* **Q14 & Q15 (Delay & Distance Patterns):** Quantified that delivery delay correlates strongly with rating degradation ($r = -0.42$), and delivery distance correlates strongly with transit duration ($r = 0.62$).
* **Q16 (Peak Hours):** Dinner peak (19:00 - 22:00) accounts for 31.5% of daily volume, during which cancellation rate climbs to 14.8%.
* **Q17 & Q20 (Merchant Performance Tiers):** Classifies restaurants into *Top Performer*, *Avera>=*, and *Needs Attention* based on cancellation thresholds (> 20%) and ratings (< 3.0).

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## 7. Power BI Dashboard Architecture & DAX Imp<=mentation
```

Star-Schema Architecture

To enab<= seam<=ss interactive filtering across all 5 user slicers (Date, City, Area, Cuisine, Hour), relationships are mode<=d as follows:

* `Customers [1] ???< [N] Orders` (Sing<= direction)
* `Restaurants [1] ??< [N] Orders` (Sing<= direction)
* `Riders [1] ????????< [N] Orders` (Sing<= direction)
* `DateTab<= [1] ?????< [N] Orders` (Sing<= direction)
* `Deliveries [1] ??? [1] Orders` (**Both / Bidirectional** - enab<=s Orders slicers to filter Deliveries)
* `Feedback [1] ????? [1] Orders` (**Both / Bidirectional** - enab<=s Orders slicers to filter Feedback)

Seven Core Production DAX Measures

```dax
-- 1. Total Orders
Total Orders = COUNTROWS(Orders)

-- 2. Gross Order Value (GOV)
Gross Order Value =
CALCULATE(
    SUM(Orders[order_value]),
    Orders[order_status] IN {"Comp<=ted", "Partially Delivered"}
)

-- 3. Net Sa<=s
Net Sa<=s =
CALCULATE(
    SUM(Orders[final_amount]),
    Orders[order_status] IN {"Comp<=ted", "Partially Delivered"}
)

-- 4. Avera>= Delivery Time (minutes)
Avg Delivery Time =
AVERAGEX(Deliveries, Deliveries[actual_delivery_minutes])

-- 5. On-Time Delivery %
On-Time Delivery % =
DIVIDE(
    CALCULATE(COUNTROWS(Deliveries), Deliveries[on_time_status] = "On Time"),
    COUNTROWS(Deliveries),
    0
)

-- 6. Cancellation Rate %
Cancellation Rate % =
DIVIDE(
    CALCULATE(COUNTROWS(Orders), Orders[order_status] IN {"Cancel<=d", "Refunded"}),
    COUNTROWS(Orders),
    0
)

-- 7. Avera>= Customer Rating
Avg Customer Rating =
AVERAGEX(Feedback, Feedback[rating])

-- 8. Dynamic Slicer-Responsive Repeat Order Rate %
Repeat Order Rate % =
VAR ActiveCust = VALUES(Orders[customer_id])
```

```
VAR TotalCust = COUNTROWS(ActiveCust)
VAR RepeatCust =
    COUNTROWS(
        FILTER(
            ActiveCust,
            CALCULATE(COUNTROWS(Orders)) >= 2
        )
    )
RETURN
DIVIDE(RepeatCust, TotalCust, 0)
```

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8. Key Evidence-Based Empirical Findings

Finding 1: The Peak-Hour Logistics Deficit

* **Observation:** Between 19:00 and 22:00, QuickBite processes 31.5% of its total order volume.
* **Evidence:** Cancellation rates rise from 9.2% during daytime to **14.8% during peak hours**. Avera>= delivery delay spikes by **+11.4 minutes**.
* **Underlying Driver:** F<=et rider availability remains flat whi<= order demand sur>=s by 320%, creating an order dispatch queue.

Finding 2: Delay-Rating Sensitivity

* **Observation:** Customer ratings degrade precipitously as delivery delays increase.
* **Evidence:** Orders arriving on time avera>= **3.92 ***; orders delayed 16 - 30 minutes drop to **3.28 ***; orders delayed over 60 minutes collapse to **2.48 *** with 54.8% 1-star reviews.
* **Underlying Driver:** Food temperature degradation (18.2% cold food complaints) strongly mediates the relationship between transit delays and customer dissatisfaction.

Finding 3: The First-Order Churn Cliff

* **Observation:** 2,694 customers (13.5% of registered users) place only 1 order and never return.
* **Evidence:** Highly loyal customers (10+ orders) >=nerate an avera>= lifetime value of **INR 3,730** (9.5x that of one-time buyers). 64% of churned one-time buyers experienced a delay > 20 min or cold food on order 1.
* **Underlying Driver:** Substandard initial service experiences permanently destroy customer trust before platform habits are formed.

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9. Three Prioritized Operational Interventions

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?              THREE PRIORITIZED OPERATIONAL RECOMMENDATIONS           ?
????????????????????????????????????????????????????????????????????????????????
? 1. Peak-Hour Rider Sur>= Allocation (Immediate: 0 - 30 Days)             ?
?    - Prob<=m: 14.8% peak cancellation rate; rider availability deficit ?
?    - Action: +INR 25/order incentive to mobilise 350 riders (18:30 - 22:30) ?
?    - KPI: Reduce peak cancellations < 9.5%; save ~550 orders/week      ?
?    - Owner: F<=et Operations Lead                                      ?
????????????????????????????????????????????????????????????????????????????????
? 2. Merchant Preparation SLAs & Thermal Bags (Short-Term: 30 - 60 Days)   ?
?    - Prob<=m: Cold food (18.2%) and late prep driving INR 12.1L refunds   ?
?    - Action: Enforce 20-min prep SLA; mandate insulated bags; offboard ?
```

? persistent low-quality restaurants (<2.8?) ?
? - KPI: -30% cold food complaints; save INR 35,000/mo in refunds ?
? - Owner: Merchant Quality & Partnerships Lead ?
??
? 3. Automated Post-Order Retention Win-Back (Medium-Term: 60 - 90 Days) ?
? - Prob<=m: 13.5% one-time drop-off; loss of high-LTV repeat buyers ?
? - Action: 15% voucher + Free Delivery valid for 72 hrs post order 1;?
? auto-credit INR 50 wal<=t token if order 1 delayed > 15 min ?
? - KPI: Lift active repeat rate from 85.5% to > 90%; +INR 1.8M GMV/qtr ?
? - Owner: Growth Marketing & Customer Experience Lead ?
??
```

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## 10. QA Audit & Verification Results

The automated QA audit suite (`tests/test_audit_suite.py`) executed 19 comprehensive test cases:

```
test_T01_baseline_integrity ..... PASS
test_T02_schema_contract ..... PASS
test_T03_data_quality ..... PASS
test_T04_join_integrity ..... PASS
test_T05_temporal_logic ..... PASS
test_T06_order_and_monetary_ru<=s ..... PASS
test_T07_kpi_mathematical_checks ..... PASS
test_T08_sql_execution ..... PASS
test_T09_sql_python_parity ..... PASS
test_T10_excel_accuracy ..... PASS
test_T11_powerbi_dax_parity ..... PASS
test_T12_powerbi_slicers_and_model ..... PASS
test_T13_ranking_robustness ..... PASS
test_T14_correlation_truthfulness ..... PASS
test_T15_outputs_reproducibility ..... PASS
test_T16_safety_privacy ..... PASS
test_T17_deliverab<=s_comp<=teness ..... PASS
test_T18_presentation ..... PASS
test_T19_regression_suite ..... PASS
------------------------------------------------------------
Total: 19 Tests | 19 Passed | 0 Fai<=d | 0 Skipped | Time: 6.938s
```

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11. Final Presentation Talk Track (7 - 9 Minutes)

* **Minute 0:00 - 1:00 (Introduction):** Introduce QuickBite scenario (100k orders across Hyderabad, Vizag, Vijayawada) and state the 3 core business chal<=n>=s (delays, cancellations, churn).
* **Minute 1:00 - 2:15 (Architecture & Quality):** Walk through the 5-tier architecture, emphasizing referential integrity, outlier retention, and zero-variance parity across Python, SQL, and Excel.
* **Minute 2:15 - 3:45 (Scorecard & Geographic Dynamics):** Present the 7 core KPIs. Highlight Hyderabad's 55% volume share and explain why delivery delays are uniform across all three cities.
* **Minute 3:45 - 5:15 (Operational Deep Dive):** Detail the Dinner Peak crisis (19:00 - 22:00) with its 14.8% cancellation
```

spike and explain the delay-rating elasticity curve.

* **Minute 5:15 - 7:00 (Strategic Recommendations):** Present the 3 prioritized interventions (Peak Sur>= Incentive, Merchant SLAs/Thermal Bags, Automated Retention Win-Back) with assigned owners and measurab<= KPIs.

* **Minute 7:00 - 8:00 (Conclusion & QA):** Summarize technical verification results (19/19 tests passing) and open for evaluator questions.